Scenario: Student Academic Performance Analysis Using Python Tools

1. Scenario Description

A college wants to analyze the academic performance of its students at the end of a semester. The college has collected information about students, including their marks in different subjects and their attendance percentages.

The college wants to identify the top-performing students, calculate average marks, compare performance across subjects, and find students who need additional academic support.

To achieve this, we use Python, NumPy, Pandas, Matplotlib, and JupyterLab to process the data, perform calculations, organize the results, and visualize the findings.

2. Data We Use

We collect the following data for each student:

Student ID	Name	Python	Java	SQL	Attendance (%)
101	Rahul	85	90	78	92
102	Priya	92	88	95	96
103	Kiran	65	70	60	75
104	Sneha	78	82	85	88
105	Arjun	55	62	58	68

Using this data, we can calculate total marks, average marks, subject-wise averages, and identify students with low marks or attendance.

3. How Each Tool Is Used

1. Python — Programming and Logic

We write the program that controls the entire analysis. Python allows us to define conditions, perform calculations, and combine different libraries.

Example use: Calculate each student's total marks and identify students whose attendance is below 75%.

How it helps: Automates the analysis and reduces manual calculations.

2. NumPy — Numerical Calculations

We use NumPy arrays to store subject marks and perform mathematical operations on them.

Example use: Calculate the average Python marks, highest Java marks, lowest SQL marks, and total marks for each student.

How it helps: Performs numerical calculations efficiently, especially when working with large datasets.

3. Pandas — Data Organization and Analysis

We use a Pandas DataFrame to organize student records into rows and columns. We can sort, filter, and summarize the information.

Example use: Find students scoring below 60 in any subject, rank students by total marks, and calculate the average marks for each subject.

How it helps: Makes structured data easier to manage, analyze, and interpret.

4. Matplotlib — Data Visualization

We create charts and graphs from the analyzed data.

Example use: Create a bar chart comparing student marks, a bar chart showing average marks by subject, and a chart comparing attendance percentages.

How it helps: Presents patterns and differences visually, making results easier for teachers and college management to understand.

5. JupyterLab — Interactive Working Environment

We write and execute Python code in separate cells, display calculation results, view DataFrames, and generate charts within the same notebook.

Example use: Create one cell to load the data, another to calculate averages, another to filter students, and another to display charts.

How it helps: Keeps the code, results, tables, and visualizations together, making the analysis easier to test, understand, and present.

4. Final Outcome

By combining these tools, the college can:

- Identify students with the highest total marks.
- Calculate average, maximum, and minimum marks.
- Compare performance across Python, Java, and SQL.
- Identify students who need academic support based on marks and attendance.
- Present the findings through charts and tables.

Tool	How We Use It	How It Helps
Python	Write the program and define the analysis logic.	Automates calculations and makes data analysis easier.
NumPy	Store marks in arrays and calculate total, average, maximum, and minimum marks.	Performs numerical calculations efficiently.
Pandas	Store student details in a DataFrame, filter students with low marks, and sort students by total marks.	Organizes, filters, and analyzes structured data easily.
Matplotlib	Create bar charts comparing student marks and graphs showing subject-wise performance.	Visualizes results and makes comparisons easier to understand.
JupyterLab	Write and execute Python code in separate cells, view results, and display charts in the same notebook.	Provides an interactive environment to perform and document the complete analysis.